

ON A SYSTEM OF TWO TETRADES OF CIRCLES; AND OTHER SYSTEMS OF TWO TETRADES.

[947]

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The investigations of the present paper were suggested to me by Mr Orr's paper, "The Contacts of certain Systems of Circles," *Proc. Camb. Phil. Soc.*, vol. VII.

1. It is possible to find *in plano* two tetrads of circles, or say four red circles and four blue circles, such that each red circle touches each blue circle: in fact, counting the constants, a circle depends upon 3 constants, or say it has a capacity = 3; the capacity of the eight circles is thus = 24; and the postulation or number of conditions to be satisfied is = 16: the resulting capacity of the system is thus *primâ facie*, $16 - 24 = 8$. It will, however, appear that, in the system considered, the true value is = 9.

2. The *primâ facie* value of the capacity being = 8, we are not at liberty to assume at pleasure three circles of the system. And, in fact, assuming at pleasure say 3 red circles, then touching each of these we have 8 circles, forming $\frac{1}{24} 8 \cdot 7 \cdot 6 \cdot 5 = 70$, tetrads of circles: taking at random any one of these tetrads for the blue circles, the remaining red circle has to be determined so as to touch each of the four blue circles, that is, by four instead of three conditions; and there is not in general any red circle satisfying these four conditions. But the 8 tangent circles do not stand to each other in a relation of symmetry, but form in fact four pairs of circles; and it is possible out of the 70 tetrads to select (and that in 6 ways) a tetrad of blue circles, such that there exists a fourth red circle touching each of these four blue circles. We have thus a system depending upon 3 arbitrary circles, and for which, therefore, the capacity is = 9. It is (as is known) possible, in quite a

different manner, out of the 70 tetrads to select (and that in 8 ways) a tetrad of blue circles such that there exists a fourth red circle touching each of these four blue circles—but the present paper relates exclusively to the first-mentioned 6 tetrads and not to these 8 tetrads.

3. I consider, in the first instance, a particular case in which the three red circles are not all of them arbitrary, but have a capacity $9 - 1 = 8$; and pass from this to the general case where the capacity is = 9. Calling the red circles 1, 2, 3 and 4; I start with the circles 1, and 2 arbitrary, and 3 a circle equal to 2: the radical axis, or common chord, of the circles 2 and 3 is thus a line bisecting at right angles the line joining the centres of the circles 2 and 3, say this is the line Ω . We have then four circles, each having its centre in the line Ω and touching the circles 1 and 2: in fact, the locus of the centre of a circle touching the circles 1 and 2 is a pair of conics, each of them having for foci the centres of these circles: the line Ω meets each of these conics in two points, and there are thus on the line Ω four points, each of them the centre of a circle touching the circles 1 and 2. But the equal circles 2 and 3 are symmetrically situate in regard to the line Ω ; and it is obvious that the four circles, having their centres on the line Ω , will each of them also touch the circle 3; we have thus the four blue circles, each of them with its centre on the line Ω and touching each of the red circles 1, 2 and 3. And it is moreover clear that, taking the red circle 4 equal to 1 and situate symmetrically therewith in regard to the line Ω , then this circle 4 will touch each of the blue circles: so that we have here the four blue circles, each of them touching the four red circles. As already mentioned, the blue circles have their centre on the line Ω , that is, the line Ω is a common orthotomic of the four blue circles.

4. By inverting in regard to an arbitrary circle we pass to the general case; the line Ω becomes thus a circle Ω , orthotomic to each of the blue circles.

Starting *ab initio*, we have here at pleasure the red circles 1, 2, 3: the circle Ω is a circle having for centre a centre of symmetry of the circles 2 and 3, and passing through the points of intersection (real or imaginary) of these two circles; the circles 2 and 3 are thus the inverses (or say the images) each of the other in regard to the circle Ω . We can then find 4 circles each of them orthotomic to Ω , and touching the circles 1 and 2: but a circle orthotomic to Ω is its own inverse or image in regard to Ω ; and it will thus touch the circle 3 which is the image of 2 in regard to Ω . We have thus the four blue circles each of them touching the red circles 1, 2 and 3; and then, taking the red circle 4 as the inverse or image of 1 in regard to Ω , this circle 4 will also touch each of the blue circles. Thus starting with the arbitrary red circles 1, 2, 3, we find the four blue circles and the remaining red circle 4, such that each of the blue circles touches each of the red circles. Since in the construction we group together at pleasure the two circles 2, 3 (out

of the three circles 1, 2, 3) and use at pleasure either of the two centres of symmetry, it appears that the number of ways in which the figure might have been completed is = 6.

5. The blue circles have a common orthotomic circle Ω , that is, the radical axis or common chord of each two of the blue circles passes through one and the same point, the centre of the circle Ω . The figure is symmetrical in regard to the red and blue circles respectively, and thus the red circles have a common orthotomic circle Ω' , that is, the radical axis or common chord of each two of the red circles passes through one and the same point, the centre of the circle Ω' .

6. Projecting stereographically on a spherical surface, the four red circles and the four blue circles become circles of the sphere; and then making the general homographic transformation, they become plane sections of a quadric surface; we have thus the theorem that on a given quadric surface it is possible to find four red sections and four blue sections such that each blue section touches each red section; and moreover the capacity of the system is = 9; viz. 3 of the red sections may be assumed at pleasure. But (as is well known) the theory of the tangency of plane sections of a quadric surface is far more simple than that of the tangency of circles: the condition in order that two sections may touch each other is simply the condition that the line of intersection of the two planes shall touch the quadric surface. And we construct, as follows, the sections touching each of three given sections: say the given sections are 1, 2, 3; through the sections 1 and 2 we have two quadric cones having for vertices say the points d_{12} and i_{12} (direct and inverse centres of the two sections): similarly, through the sections 1 and 3 we have two quadric cones vertices d_{13} and i_{13} respectively, and through the sections 2 and 3 we have two quadric cones vertices d_{23} and i_{23} respectively; the points $d_{12}, i_{12}, d_{13}, i_{13}, d_{23}, i_{23}$, lie three and three in four intersecting lines or axes, viz. these are $d_{23}d_{31}d_{12}$, $d_{23}i_{31}i_{12}$, $d_{31}i_{12}i_{23}$, $d_{12}i_{23}i_{31}$ respectively. Through any one of these axes, say $d_{23}d_{31}d_{12}$, we may draw to the quadric surface two tangent planes each touching the three cones which have their vertices in the points d_{23}, d_{31}, d_{12} respectively; and the section by either tangent plane is thus a section touching each of the three given sections 1, 2, 3; we have thus the eight tangent sections of these three sections.

7. Taking as three of the red sections the arbitrary sections 1, 2, 3; and grouping together two at pleasure of these sections, say 2 and 3; we may take for the blue sections the two sections through the axis $d_{23}d_{31}d_{12}$, and those through the axis $d_{23}i_{31}i_{12}$; we have thus the four blue sections touching each of the given red sections 1, 2, 3; and this being so, there exists a remaining red section 4 touching each of the blue sections; we have thus the four blue sections touching each of the red sections 1, 2, 3 and 4. This implies that the vertices or points d_{24} and d_{34} lie on the axis $d_{23}d_{31}d_{12}$, and that the vertices or points i_{24} and i_{34} lie on the axis $d_{23}i_{31}i_{12}$; or, what is the same thing, that the four sections 1, 2, 3, 4 have in common an axis $d_{23}d_{31}d_{12}d_{24}d_{34}$ and also an axis $d_{23}i_{31}i_{12}i_{24}i_{34}$.

8. If the quadric surface be a flat surface (*surface aplatie*) or conic, then the red sections become chords of the conic; the axes are lines in the plane of the conic, and thus the tangent planes through an axis each coincide with the plane of the conic, and it would at first sight appear that any theorem as to tangency becomes nugatory. But this is not so; comparing with the last preceding paragraph, we still have the theorem: on a given conic, taking at pleasure any three chords

1, 2, 3, it is possible to find a fourth chord 4, such that the four chords have in common an axis $d_{23}d_{31}d_{12}d_{24}d_{34}$ and also an axis $d_{23}i_{31}i_{12}i_{24}i_{34}$. And the analytical theory (although somewhat complex) is extremely interesting. Considering the conic $xz - y^2 = 0$, the coordinates of a point on the conic are given by $x : y : z = 1 : \theta : \theta^2$, or say any point of the conic is determined by its parameter θ ; and this being so, considering any three chords 1, 2, 3, I take for the two extremities of 1 the values ϵ, ζ ; for those of 2 the values α, β ; and for those of 3 the values γ, δ ; the remaining chord 4 is to be determined as above, and I take for its two extremities the values E, Z .

9. Starting with the chords 1, 2, 3, we have each of the points d_{23} , &c., as the intersection of two lines, viz. these are

$$\begin{array}{ll} \text{for } d_{23}: & \begin{cases} \alpha\beta z - y(\alpha + \beta) + x = 0, \\ \gamma\delta z - y(\gamma + \delta) + x = 0; \end{cases} & \text{for } d_{31}: & \begin{cases} \alpha\gamma z - y(\alpha + \gamma) + x = 0, \\ \beta\delta z - y(\beta + \delta) + x = 0; \end{cases} \\ \text{for } d_{12}: & \begin{cases} \alpha\epsilon z - y(\alpha + \epsilon) + x = 0, \\ \beta\zeta z - y(\beta + \zeta) + x = 0; \end{cases} & \text{for } i_{12}: & \begin{cases} \alpha\zeta z - y(\alpha + \zeta) + x = 0, \\ \beta\epsilon z - y(\beta + \epsilon) + x = 0; \end{cases} \end{array}$$

and similarly for i_{23}, i_{31} . We thence find without difficulty for the axis $d_{23}d_{31}d_{12}$ the equation

$$x \{ \alpha \beta (\delta \epsilon - \gamma \zeta) + \gamma \delta (\zeta \alpha - \epsilon \beta) + \epsilon \zeta (\beta \gamma - \alpha \delta) \} + y \{ (\beta - \alpha)(\gamma \delta - \epsilon \zeta) + (\delta - \gamma)(\epsilon \zeta - \alpha \beta) + (\zeta - \epsilon)(\alpha \beta - \gamma \delta) \} + z \{ -(\delta \epsilon - \gamma \zeta) - (\zeta \alpha - \epsilon \beta) - (\beta \gamma - \alpha \delta) \} = 0;$$

the equation of the axis $d_{23}i_{31}i_{12}$ is obtained herefrom by the interchange of ϵ and ζ .

10. The points d_{24} and d_{34} will lie upon the first-mentioned axis if only d_{34} lies upon this axis, viz. if we have

$$[\alpha \beta (\delta \epsilon - \gamma \zeta) + \gamma \delta (\zeta \alpha - \epsilon \beta) + \epsilon \zeta (\beta \gamma - \alpha \delta)] (\beta + E - \alpha - Z) + [(\beta - \alpha)(\gamma \delta - \epsilon \zeta) + (\delta - \gamma)(\epsilon \zeta - \alpha \beta) + (\zeta - \epsilon)(\alpha \beta - \gamma \delta)] (\beta E - \alpha Z) + [-(\delta \epsilon - \gamma \zeta) - (\zeta \alpha - \epsilon \beta) - (\beta \gamma - \alpha \delta)] [\dots] = 0.$$

Reducing this equation, the factor $\alpha - \beta$ divides out, and we finally obtain

$$A + BE + CZ + DEZ = 0;$$

say this is

$$A + BE + CZ + DEZ = 0,$$

where

$$\begin{aligned} A &= \cdot, \\ B &= -(\gamma - \alpha) \beta \delta \cdot + (\alpha \beta - \gamma \delta) \zeta + (\beta - \delta) \epsilon \zeta, \\ C &= -(\beta - \delta) \alpha \gamma + (\alpha \beta - \gamma \delta) \epsilon + (\gamma - \alpha) \epsilon \zeta, \\ D &= -(\alpha \delta - \beta \gamma) - (\beta - \delta) \epsilon - (\gamma - \alpha) \zeta, \end{aligned}$$

viz. this is the condition for the existence of the axis $d_{23}d_{31}d_{12}d_{24}d_{34}$.

We interchange herein ϵ, ζ and also E, Z , and we thus obtain

$$A' + C'E + B'Z + D'EZ = 0,$$

where

$$\begin{aligned} A' &= \cdot, \\ B' &= \cdot + (\beta - \delta) \alpha \gamma \zeta + (\gamma - \alpha) \beta \delta \epsilon + (\alpha \beta - \gamma \delta) \epsilon \zeta, \\ C' &= (\alpha \beta - \gamma \delta) \epsilon + (\zeta - \delta) \cdot, \\ D' &= \cdot + (\alpha \beta - \gamma \delta) \zeta + (\gamma - \alpha) \cdot, \end{aligned}$$

viz. this is the condition for the existence of the axis $d_{23}i_{31}i_{12}i_{24}i_{34}$.

11. I remark that we have

$$\begin{aligned} A' &= A + a(\epsilon - \zeta), & B' &= B + b(\epsilon - \zeta), \\ C' &= C + c(\epsilon - \zeta), & D' &= D + d(\epsilon - \zeta), \end{aligned}$$

where

$$\begin{aligned} a &= (\beta - \delta) \alpha \gamma - (\gamma - \alpha) \beta \delta = \alpha \beta (\gamma + \delta) - \gamma \delta (\alpha + \beta), \\ b &= c = \gamma \delta - \alpha \beta, \\ d &= \alpha + \beta - \gamma - \delta, \end{aligned}$$

and further that

$$B - C = \alpha \beta (\gamma + \delta) - \gamma \delta (\alpha + \beta) + (\gamma \delta - \alpha \beta) (\epsilon + \zeta) + (\alpha + \beta - \gamma - \delta) \epsilon \zeta;$$

say this is II.

12. It thus appears that, for the determination of E, Z , we have

$$\begin{aligned} A + BE + CZ + DEZ &= 0, \\ A' + C'E + B'Z + D'EZ &= 0. \end{aligned}$$

Eliminating Z , we find

$$\frac{A + BE}{A' + C'E} = \frac{C + DE}{B' + D'E},$$

that is,

$$(AC' - A'C) + (AD' - A'D + BB' - CC')E + (BD' - B'D)E^2 = 0;$$

upon reducing the coefficients of this equation it appears that they contain each of them the factor Π , and throwing out this factor, the equation is

$$\epsilon [\alpha\beta(\gamma - \delta) + \gamma\delta(\beta - \alpha) + (\alpha\delta - \beta\gamma)\zeta] + \cdots = 0;$$

this contains obviously the factor $E - \epsilon$, or throwing out this factor, we have for E the simple equation

$$[\alpha\beta(\gamma - \delta) + \gamma\delta(\beta - \alpha)] + (\alpha\delta - \beta\gamma)(E + \zeta) + (\beta - \alpha + \gamma - \delta)\zeta E = 0.$$

In a similar manner it may be shown that the two equations give for Z the like simple equation

$$[\alpha\beta(\gamma - \delta) + \gamma\delta(\beta - \alpha)] + (\alpha\delta - \beta\gamma)(Z + \epsilon) + (\beta - \alpha + \gamma - \delta)\epsilon Z = 0,$$

viz. starting from the chords 1, 2, 3 which depend on the parameters (ϵ, ζ) , (α, β) , (γ, δ) respectively, these last two equations give the parameters (E, Z) of the chord 4.

REPORT OF A COMMITTEE APPOINTED FOR THE PURPOSE OF
CARRYING ON THE TABLES CONNECTED WITH THE PELLIAN EQUATION
FROM THE POINT WHERE THE WORK WAS LEFT BY DEGEN IN 1817.

[948]

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We have, on the Pellian Equation, Degen’s tables, the title of which is “Canon Pellianus sive Tabula simplicissimam æquationis celebratissimæ $y^2 = ax^2 + 1$ solutionem pro singulis numeri dati valoribus ab 1 usque ad 1000 in numeris rationalibus iisdemque integris exhibens.” Autore Carolo Ferdinando Degen. Hafniæ, apud Gerhardum Bonnierum, MDCCCXVII., 8vo. Introductio, pp. v—xxiv. Tabula I. Solutionem æquationis $y^2 - ax^2 - 1 = 0$ exhibens, pp. 3—106. Tabula II. Solutionem æquationis $y^2 - ax^2 + 1 = 0$, quotiescunque valor ipsius a talem admiserit, exhibens, pp. 109—112.

The mode of calculation is explained in the Introduction, and illustrated by the examples of the numbers 209, 173.

As to the first of these, the entry in Table I. is

209	14, 2, 5, 3, (2)
	1, 13, 5, 8, 11
	3220
	46551

where the first line gives the expression of $\sqrt{209}$ as a continued fraction, viz. we have

$$\sqrt{209} = 14 + \frac{1}{2 + \frac{1}{5 + \frac{1}{3 + \frac{1}{2 + \frac{1}{3 + \frac{1}{5 + \frac{1}{2 + \frac{1}{28 + \frac{1}{2 + \dots}}}}}}}} \&c.$$

the denominators being 2, 5, 3, (2), 3, 5, 2, then 28, which is the double of the integer part 14, and then again 2, 5, 3, (2), 3, 5, 2, and so on, the parentheses of the (2) being used to indicate that this is the middle term of the period.

The second row gives auxiliary numbers occurring in the calculation of the first row and having a meaning, as will presently appear. Observe that the 11 which comes under the (2) should also be printed in parentheses (11), but this is not done.

The process for the calculation of the x, y is as follows:

209			
	1	0	+1
14	14	1	−13
2	29	2	+5
5	159	11	−8
3	506	35	+(11)
(2)	1171	81	−13
3	4019	278	+5
5	21266	1471	−8
2		3220	
28	46551		+1

viz. writing down as a first column the numbers of the first row, and beginning the second column with 1, 14 (14 the number at the head of the first column), and the third column with 0, 1, we calculate the numbers of the second column, $29 = 2 \cdot 14 + 1$, $159 = 5 \cdot 29 + 14$, $506 = 3 \cdot 159 + 29$, &c., and the numbers of the third column in like manner, $2 = 2 \cdot 1 + 0$, $11 = 5 \cdot 2 + 1$, $35 = 3 \cdot 11 + 2$, &c.; and then writing down as a fourth column the numbers of the second row with the signs +, − alternately, we have a series of equations $y^2 - ax^2 = \pm A$, viz.

$$\begin{aligned} 1^2 - 209 \cdot 0^2 &= +1, \\ 14^2 - 209 \cdot 1^2 &= -13, \\ 29^2 - 209 \cdot 2^2 &= +5, \end{aligned}$$

the last of them being

$$(46551)^2 - 209\,(3220)^2 = +1,$$

this last corresponding as above to the value +1, and the numbers 46551 and 3220 being accordingly the y and x given in the fourth and third rows of the table.

As to the second of the foregoing numbers, 173, the only difference is that the period has a double middle term, viz. the entry in the Table I. is

173

13, 6, (1, 1)

1, 4, (13, 13)

190060

2499849

The first row gives the expression of $\sqrt{173}$, viz. that is

$$\sqrt{173} = 13 + \frac{1}{6 + \frac{1}{1 + \frac{1}{1 + \frac{1}{6 + \frac{1}{26 + \dots}}}}} \&c.$$

the denominators being 6, 1, 1, 6, then 26 (the double of the integer part 13), and then again 6, 1, 1, 6, and so on. In the second row I remark that Degen prints the parentheses (13, 13) for the double middle term.

The process for the calculation of the x, y is similar to that in the former case, viz. we have

173			
	1	0	+1
13	13	1	−4
6	79	6	+13
(1, 1)	92	7	−1
	171	13	+13
6			
26	1118	85	+4

where the second and third columns begin 1, 13 and 0, 1 respectively, and the remaining terms are calculated $79 = 6 \cdot 13 + 1$, $92 = 1 \cdot 79 + 13$, &c., and $6 = 6 \cdot 1 + 0$, $7 = 1 \cdot 6 + 1$, &c.; and then writing down as a fourth column the terms of the second row with the signs +, − alternately, we have

$$1^2 - 173 \cdot 0^2 = +1,$$
$$13^2 - 173 \cdot 1^2 = -4,$$
$$79^2 - 173 \cdot 6^2 = +13,$$

the last equation being

$$(1118)^2 - 173\,(85)^2 = -1,$$

the term for the last equation being always in a case such as the present one, not +1, but −1. The final numbers 1118, 85 are consequently entered not in Table I., but in Table II., viz. the entry in this table is

173

85

1118

and thence we calculate the numbers y, x of Table I., viz. these are

$$2499849 = 2 \cdot (1118)^2 + 1,$$
$$190060 = 2 \cdot 1118 \cdot 85.$$

Generally Table II. gives for each value of a , comprised therein, values of x, y , such that $y^2 = ax^2 - 1$, and then writing $y_1 = 2y^2 + 1$, $x_1 = 2xy$, we have

$$y_1^2 = (2ax^2 - 1)^2 = 4a^2x^4 - 4ax^2 + 1 = a \cdot 4x^2 (ax^2 - 1) + 1 = ax_1^2 + 1,$$

so that x_1, y_1 are for the same value of a the values of x, y in Table I.

It is to be remarked that the heading of Table II. is not perfectly accurate, for it purports to give for every value of a , for which a solution exists, a solution of the equation $y^2 = ax^2 - 1$. What it really gives is the solution for each value of a for which the period has a double middle term. But if $a = \alpha^2 + 1$, then obviously we have a solution $y = \alpha, x = 1$, and for any such value of a the period has a single middle term, viz. the entry in Table I. is

$$\alpha^2 + 1 \left| \begin{array}{l} \alpha, \ (2\alpha) \\ 1, \ 1 \\ 2\alpha \\ 2\alpha^2 + 1 \end{array} \right.$$

and we, in fact, have

$\alpha^2 + 1$			
	1	0	-1
α	α	1	+1
(2α)	$2\alpha^2 + 1$	2α	+1

that is,

$$\begin{aligned} 1^2 - (\alpha^2 + 1) \cdot 0^2 &= +1, \\ \alpha^2 - (\alpha^2 + 1) \cdot 1^2 &= -1, \\ (2\alpha^2 + 1)^2 - (\alpha^2 + 1) (2\alpha)^2 &= +1. \end{aligned}$$

The foregoing instances of the calculation of x, y in the case of the numbers 209 and 173 suggest a table which may be regarded as an extended form of Degen’s tables; viz. such a table, from $a = 2$ to $a = 99$, is as follows:

SPECIMEN OF EXTENDED FORM OF TABLE IN REGARD TO THE PELLIAN EQUATION.

a	y	x	$y^2 - ax^2$
2	1	0	+1
	1	1	−1
(2)	3	2	+1
3	1	0	+1
	1	1	−2
(1)	2	1	+1
2			
5	2	1	+1
	2	1	−1
(4)	9	4	+1
4			
6	2	1	+1
	2	1	−2
(2)	5	2	+1
4			
7	2	1	+1
1	3	1	−3
(1)	5	2	+2
1	8	3	−3
4			+1
8	2	1	+1
	3	1	−4
(1)	17	6	+1
4			
10	3	1	+1
	3	1	−1
(6)	19	6	+1
6			
11	3	1	+1
	3	1	−2
(3)	10	3	+1
6			
12	3	1	+1
	3	1	−3
(2)	7	2	+1
6			
13	3	1	+1
	4	1	−3
1	7	2	−5
1	11	3	+4
1	18	5	+1
(6)			
14	3	1	+1
	4	1	−1
(2)	11	3	−2
6	15	4	+1
15	3	1	+1
(1)	4	1	+1
6			
17	4	1	+1
(8)	8	2	−1
8	33	8	+1
18	4	1	+1
(4)	9	2	−2
2	17	4	−1

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—CONTINUED.

a	y	x	$y^2 - ax^2$
21	4	1	+1
	4	1	−3
1	5	1	+4
1	9	2	−5
(2)	23	5	+4
1	32	7	−5
1	55	12	+1
8			
22	4	1	+1
1	4	1	−2
2	5	1	+3
(4)	14	3	+3
2	61	13	+3
1	136	29	−6
8	197	42	+1
23	4	1	+1
1	4	1	−7
(3)	5	1	+2
1	19	4	−7
8	24	5	+1
24	4	1	+1
(1)	4	1	−1
8	5	1	+1
26	5	1	+1
(10)	5	1	−1
10	51	10	+1
27	5	1	+1
(5)	5	1	−2
10	26	5	+1
28	5	1	+1
3	5	1	−3
(2)	16	3	+4
3	37	7	−3
10	127	24	+1
29	5	1	+1
2	5	1	−4
(1)	11	2	+5
2	16	3	−5
10	27	5	+4
	70	13	−1
	727	135	+1
30	5	1	+1
(2)	5	1	−5
10	11	2	+1
31	5	1	+1
1	5	1	−6
1	6	1	+5
3	11	2	−3
(5)	39	7	+2
3	206	37	−3
1	657	118	+5
1	863	155	−6
10	1520	273	+1
32	5	1	+1
1	5	1	−7

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—CONTINUED.

a	y	x	$y^2 - ax^2$
35	5	1	+1
(1)	6	1	−1
10			+1
37	6	1	+1
(12)			
12	73	12	+1
38	6	1	+1
(6)			
12	37	6	+1
39	6	1	+1
(4)			
12	25	4	+1
40	6	1	+1
(3)	6	1	−4
12	19	3	+1
41	6	1	+1
(2)	6	1	−5
12	13	2	+5
	32	5	+1
42	6	1	+1
(2)	6	1	−6
12	13	2	+1
43	6	1	+1
1	6	1	−7
1	7	1	+6
3	13	2	−3
1	46	7	+9
(5)	59	9	−2
1	341	52	+9
3	400	61	−7
1	1541	235	+6
1	1941	296	−7
12	3482	531	+1
44	6	1	+1
	6	1	−8
1	7	1	+5
1	13	2	−7
1	20	3	+4
(2)	53	8	−5
1	73	11	+8
1	126	19	−5
12	199	30	+1
45	6	1	+1
	6	1	−9
(2)	7	1	+4
2	20	3	−9
12	47	7	+4
	114	17	−9
	161	24	+4
46	6	1	+1
1	6	1	−10
3	7	1	+3
1	27	4	−7
1	34	5	+6
2	61	9	−5

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—CONTINUED.

a	y	x	$y^2 - ax^2$
50	7	1	+1
(14)	7	1	−1
14	99	14	+1
51	7	1	+1
(7)	7	1	−2
14	50	7	+1
52	7	1	+1
4	7	1	−3
1	29	4	−4
(2)	36	5	+9
1	101	14	−3
4	137	19	+9
14	649	90	+1
53	7	1	+1
3	7	1	−4
(1)	22	3	+7
3	29	4	−7
14	51	7	+4
	182	25	−5
54	7	1	+1
2	7	1	−5
1	15	2	+2
(6)	22	3	−3
1	147	20	+9
2	169	23	−1
14	485	66	+9
55	7	1	+1
2	7	1	−6
(2)	15	2	+5
2	37	5	−6
14	89	12	+1
56	7	1	+1
(2)	7	1	−7
14	15	2	+1
57	7	1	+1
1	7	1	−8
1	8	1	+7
(4)	15	2	−3
1	68	9	+7
1	83	11	−8
14	151	20	+1
58	7	1	+1
1	7	1	−9
1	8	1	+6
(1)	15	2	−7
1	23	3	+9
1	38	5	−10
14	99	13	+1
59	7	1	+1
1	7	1	−10
2	8	1	+5
(7)	23	3	−5
2	169	22	−11
1	361	47	−10
14	520	60	+1

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—CONTINUED.

a	y	x	$y^2 - ax^2$
	1523	195	+4
	5639	722	−3
1	24079	3083	+12
14	29718	3805	−13
62	7	1	+1
1	7	1	−13
(6)	8	1	+2
1	55	7	−7
14	63	8	+1
63	7	1	+1
(1)	7	1	−14
14	8	1	+1
65	8	1	+1
(16)	8	1	−1
16	129	16	+1
66	8	1	+1
(8)	8	1	−2
16	65	8	+1
67	8	1	+1
5	8	1	−3
2	41	5	−6
1	90	11	+7
1	131	16	−9
(7)	221	27	+9
1	1678	205	+9
1	1899	232	+7
2	3577	437	−6
5	9053	1106	+3
16	48842	5967	+1
68	8	1	+1
(4)	8	1	−4
16	33	4	+1
69	8	1	+1
3	8	1	−5
1	25	3	−5
1	83	10	+4
(4)	108	13	−5
1	515	62	+4
3	623	75	−5
3	2384	297	−6
16	7775	936	+1
70	8	1	+1
2	8	1	−6
1	17	2	+3
(2)	25	3	−7
1	67	8	+9
2	92	11	−6
16	251	30	−11
71	8	1	+1
2	8	1	−7
2	17	2	+2
1	42	5	−11
(7)	59	7	+2
2	455	54	−9
2	514	61	+5

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—CONTINUED.

a	y	x	$y^2 - ax^2$
(1)	9	1	+7
1	17	2	−7
16	26	3	+10
	43	5	−7
75	8	1	+1
(1)	8	1	−11
1	9	1	+6
16	17	2	−11
	26	3	+1
76	8	1	+1
1	8	1	−12
1	9	1	+5
2	26	3	−8
1	35	4	+9
1	61	7	−3
5	340	39	+4
(4)	1421	163	−3
5	7445	854	−12
1	8866	1017	+5
1	16311	1871	+1
2	41488	4759	+1
16	57799	6630	+1
77	8	1	+1
3	9	1	+4
(2)	35	4	−13
3	79	9	+4
1	272	31	−1
16	351	40	+1
78	8	1	+1
1	8	1	−14
(4)	9	1	+3
1	44	5	−14
16	53	6	+1
79	8	1	+1
1	8	1	−15
(7)	9	1	+2
1	71	8	−15
16	80	9	+1
80	8	1	+1
(1)	8	1	−16
16	9	1	+1
82	9	1	+1
(18)	9	1	−1
18	163	18	+1
83	9	1	+1
(9)	9	1	−2
18	82	9	+1
84	9	1	+1
(6)	9	1	−3
18	55	6	+1
85	9	1	+1
4	9	1	−4
(1)	37	4	+9
1	46	5	−5
4	82	9	−7

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—CONTINUED.

a	y	x	$y^2 - ax^2$
87	9	1	+1
(3)	9	1	−6
18	28	3	+1
88	9	1	+1
2	9	1	−7
1	19	2	+9
(1)	28	3	−8
1	47	5	+9
2	75	8	−7
18	197	21	+1
89	9	1	+1
2	9	1	−8
(3)	19	2	+5
2	66	7	−5
18	217	23	+8
	500	53	+1
90	9	1	+1
(2)	9	1	−9
18	19	2	+1
91	9	1	+1
1	9	1	−10
1	10	1	+9
5	19	2	−3
(1)	105	11	−6
5	124	13	+5
1	725	76	−9
1	849	89	+10
18	1574	165	+1
92	9	1	+1
1	9	1	−11
1	10	1	+8
2	19	2	−7
(4)	48	5	−11
2	211	22	+5
1	470	49	−7
1	681	71	+8
18	1151	120	−11
93	9	1	+1
1	9	1	−12
1	10	1	+7
1	19	2	−11
4	29	3	+4
(6)	135	14	−9
4	839	87	−4
1	3491	362	+5
1	4330	449	−11
1	7821	811	+5
18	12151	1260	−12
94	9	1	+1
1	9	1	−13
2	10	1	+6
3	29	3	−13
1	97	10	+9
1	126	13	−15
5	222	22	+2

SPECIMEN OF EXTENDED FORM OF PELLIAN EQUATION TABLE—CONTINUED.

<i>a</i>	<i>y</i>	<i>x</i>	$y^2 - ax^2$
97	9	1	+1
1	9	1	−16
5	10	1	+3
1	59	6	−13
1	69	7	+8
(1)	128	13	−9
1	197	20	+9
1	325	33	−8
5	522	53	+13
1	847	86	−3
1	4757	483	+16
18	5604	569	+1
98	9	1	+1
(8)	9	1	−17
18	10	1	+2
	89	9	−17
	99	10	+1
99	9	1	+1
(1)	9	1	−18
18	10	1	+1

The meaning hardly requires explanation; for each number *a*, we have a series of pairs of increasing numbers, *y*, *x*, satisfying a series of equations $y^2 = ax^2 \pm b$; thus

$$a = 14$$

<i>y</i>	<i>x</i>	$y^2 - ax^2$
1	0	$1 - 14 \cdot 0 = 1,$
3	1	$9 - 14 \cdot 1 = -5,$
4	1	$16 - 14 \cdot 1 = +2,$
11	3	$121 - 14 \cdot 9 = -5,$
15	4	$225 - 14 \cdot 16 = +1.$

The following table, calculated under the superintendence of the Committee, extends from *a* = 1001 to *a* = 1500 (square numbers omitted); it is (with slight typographical variations) nearly but not exactly in the form of Degen’s Table I., the chief difference being that for a number *a* having a double middle term, or of the form $\alpha^2 + 1$ (such number being further distinguished by an asterisk), the *x*, *y* entered in the table are the solutions, not of the equation $y^2 = ax^2 + 1$, but of the equation $y^2 = ax^2 - 1$. As remarked above, if we have $y^2 = ax^2 - 1$, then writing $y_1 = 2y^2 + 1$ and $x_1 = 2xy$, we obtain $y_1^2 = ax_1^2 + 1$.

Moreover, for each value of *a*, in the first line, the first term, which is the integer part of $\sqrt{a}$, is separated from the others by a semicolon, and the 1, which is the corresponding first term of the second line, is omitted.

The calculations were made by C. E. Bickmore, M.A., of New College, Oxford: his values for *x* and *y* have been revised as presently mentioned, but it has been assumed that his values for the periods and subsidiary numbers (forming the first and second lines of each division of the table) are accurate; in fact, any error therein would cause the resulting values of *x* and *y* to be wildly erroneous; but (except in a single instance which was accounted for) the errors in *x* and *y* were in every case in a single figure or two or three figures only.

The values of *x* and *y* were in every case examined by substitution in the equation ($y^2 = ax^2 + 1$, or $y^2 = ax^2 - 1$, as the case may be), which should be satisfied by them. These verifications were for the most part made by A. Graham, M.A., of the Observatory, Cambridge. As already mentioned, some errors were detected, and these have been, of course, corrected. The values of *x*, *y* given in the table thus satisfy

in every case the proper equation $y^2 = ax^2 + 1$, or $y^2 = ax^2 - 1$; on the ground above referred to, it is believed that the periods and subsidiary numbers are also accurate.

It may be remarked, in regard to the verification of the equation $y^2 = ax^2 \pm 1$ for large values of a and y , it is in practice easier and safer to calculate $ax^2 \pm 1$, and then to compare the square root thereof with the given value of y , than to further calculate the value of y^2 .

THE TABLE 1001 TO 1500.

The following table lists, for each integer a from 1001 to 1500 (perfect squares omitted), the continued-fraction expansion of $\sqrt{a}$, the auxiliary period-numbers, and the resulting fundamental solution (x, y) of the Pellian equation. Entries marked by an asterisk $*$ correspond to a double middle term; in those cases the tabulated (x, y) satisfy $y^2 = ax^2 - 1$. A semicolon separates the integer part from the period; parenthesised numbers denote middle terms.

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
1001	31; 1, 1, 1, 3, 3, 2, (4); 40, 23, 32, 16, 17, 25, (13)	33532 10 60905
1002	31; 1, 1, 1, 8, 2, 1, 1, 1, 3, (10); 41, 22, 39, 7, 23, 31, 26, 33, 17, (6)	65 35248 2068 69247
1003	31; 1, 2, (31); 42, 21, (2)	285 9026
1004	31; 1, 2, 5, 2, 2, 1, 7, 4, 1, 2, 1, 11, 1, (14); 43, 20, 11, 25, 19, 41, 8, 13, 40, 17, 44, 5, 55, (4)	85 24164 59730 2700 96330 24199
1005	31; 1, 2, 2, 1, 5, 15, 1, 2, (12); 44, 19, 20, 39, 11, 4, 41, 21, (5)	930 59568 29501 49761
1006	31; 1, 2, 1, 1, 5, 1, 3, 2, 1, 1, 1, 1, 1, 9, 1, 20, 4, 5, 1, 1, 12, 6, 1, (30); 45, 18, 29, 33, 10, 43, 15, 22, 31, 27, 30, 25, 37, 6, 55, 3, 15, 11, 30, 33, 5, 9, 53, (2)	4 45346 14025 55749 21748 141 25267 56378 02146 05455
1007	31; 1, 2, (1); 46, 17, (38)	127 17
1008	31; 1, (2); 47, (16)	4 127
1009*	31; 1, (3, 3); 48, (15, 15)	17 540
1010*	31; 1, 3, (1, 1); 49, 14, (31, 31)	265 1303
1011	31; 1, 3, 1, (9); 50, 13, 47, (6)	41 8426
1012	31; 1, 4, 3, 6, 1, 3, 8, 1, (4); 51, 12, 19, 9, 43, 16, 7, 48, (11)	1013 02110 32226 17399
1013*	31; 1, 4, 1, 4, 15, 1, 2, (2, 2); 52, 11, 44, 13, 4, 43, 19, (23, 23)	123 52985 3931 66618
1014	31; 1, 5, 2, 1, 1, 1, 1, (20); 53, 10, 23, 30, 29, 25, 38, (3)	1 46266 46 56965
1015	31; 1, 6, 10, (2); 54, 9, 6, (29)	11076

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
		3 52871
1016	31; 1, (6); 55, (8)	2558 9 09655 84992
1017	31; 56, 7, 8, 49, 9, 13, 41, 16, 11, 31, 32, (9)	290 09322 97217 27 28333
1018*	31; 1, 9, 1, 1, 1, 6, 2, (3, 3); 57, 6, 39, 23, 38, 9, 26, (17, 17)	870 50499 19 07764 36539
1019	31; 58, 5, 47, 14, 35, 25, 34, 17, 7, 49, 10, 29, (2)	608 99233 21730 16
1020	31; 1, (14); 59, (4)	511 98 65001 29666 69564 06909
1021*	31; 1, 20, 3, 6, 1, 3, 2, 1, 1, 12, 5, 4, 15, 1, 2, 1, 4, 1, 1, 2, 1, 1, 1, (5, 31); 60, 3, 20, 9, 44, 15, 23, 27, 36, 5, 12, 15, 4, 45, 17, 41, 12, 33, 29, 20, 33, 25, 36, (11, 11)	52 17280 37258 48825 15030 1023

TABLE 1001 TO 1500—CONTINUED.

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
1022	31; 1, (30); 61, (2)	32 1023
1023	31; (1); (62)	1 32
1025*	32; (64); (1)	1 32
1026	32; (32); (2)	32 1025
1027	32; 21, 1, 1, 6, 2, (4); 3, 22, 39, 9, 27, (13)	41 54868 1331 50393
1028	32; (16); (4)	16 513
1029	32; 5, 49, 12, 25, 20, 37, 17, 4, 47, 15, 20, 43, (3)	303 57068 95884 9737 94964 66615
1030	32; 6, 41, 21, 9, 45, 14, 39, 19, 31, 30, 21, 26, (5)	2 75531 87434 88 43070 11291
1031	32; 7, 10, 19, 13, 35, 26, 31, 25, 38, 5, 14, 29, 35, (2)	2029 75370 82877 65173 74486 64200
1032	32; (8); (8)	2578 82780

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
		49654 25620
1050	32; 2, 1, 25, (10); 26, 25, (6)	270 8749
1051	32; 2, 17, 1, 1, 14, 15, 6, 35, 29, 18, 5, 42, 21, 10, 9, 3, 30, 7, 46, 15, 25, 19, 4, 5, (2); 27, 2, 1, 30, 23, 4, 4, 10, 1, 1, 3, 12, 1, 2, 6, 7, 21, 2, 8, 1, 3, 2, 2, 1, (31)	11935 20 2 94697 22410 66655 86570 73250
1052	32; 28, 19, 3, 7, 1, 4, 9, (16); 3, 19, 8, 35, 13, 7, (4)	3 00767 87132 79112
1053	32; 29, 2, 3, (4); (13)	649 30
1054	32; 30, 9, 62, 1, 2, 1, 1, 4, 2, 2, 1, 1, 1, 3, 1, 2, 3, 4, (32); 2, 19, 50, 33, 13, 25, 21, 33, 26, 35, 15, 38, 21, 18, 15, (2)	10335 86450 05416 3 35557 62560 56895
1055	32; (12); (5)	1689 52
1056	32; 1, (5); (32)	2 65
1057	32; 21, 5, 1, 6, 2, 1, 1, 3, 2, 7, 1, 2, 4, 1, 1, 1, (8); 33, 1, 3, 11, 48, 9, 24, 29, 33, 16, 27, 8, 41, 21, 13, 37, 24, 39, (7)	32992 7 37084
1058	32; 1, 8, 1, 3, 1, 3, (32); 34, 7, 1, 47, 14, 41, 17, (2)	95685 1318 36323
1059	32; 1, 31, 11, 25, 2, 1, 2, 3, 1, (31); 35, 1, 32, 5, 2, 23, 21, 35, 22, 15, 49, (2)	53146 68081
1060	32; 1, 9, 5, 7, 3, 44, 1, 11, 1, 15, 9, (16); 36, 1, 28, 2, 1, 12, 3, 5, 1, 1, 2, 15, 1, 8, 2, 1, 2, (1, 1)	38369 737 64856
1061*	32; 1, 9, 5, 7, 3, 44, 1, 11, 1, 15, 9, (16); 37, 27, 49, 44, 5, 20, 11, 35, 28, 25, 4, 55, 7, 23, 35, 20, (31, 31)	82899 11530
1062	32; 1, 2, 3, 23, 19, (32); 38, (2)	3 06917 9 24122 86971 91521

TABLE 1001 TO 1500—CONTINUED.

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
1063	32; 1, 1, 26, 25, 27, 37, 1, 10, 3, 1, 2, 1, 6, 1, 1, 21, 4, 1, (31); 39, 6, 17, 39, 18, 43, 9, 31, 34, 3, 13, 51, (2)	12685 5353 15274
1064	32; 1, 1, (28); 40, 24, 31, (1)	21 685
1065	32; 1, 1, 23, 1, 2, 1, 3, 2, 1, 5, (4); 41, 23, 1, 35, 19, 39, 16, 21, 40, 11, (15)	33160

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
		826 67999
1066*	32; 1, 2, 2, 1, 5, 1, 6, (2, 2); 42, 39, 10, 49, 9, (25, 25)	1 34907 05205
1067	32; 1, (22); 43, 21, (1)	3 34
1068	32; 1, 8, 49, 11, 5, (16); 44, 7, 1, (4)	53094 3
1069*	32; 1, 2, 3, 20, 17, 29, 31, 1, 21, 4, 3, 5, 7, 12, 1, 15, 2, 2, 1, 3, 1, 1, 1, 4, 1, (4, 4); 45, 6, 3, 15, 19, 12, 9, 5, 57, 4, 27, 20, 39, 15, 36, 25, 37, 12, 45, (13, 13)	115 39207 186 40986 37841 77726
1070	32; 1, 2, 5, 26, 11, 31, 34, (5); 46, 1, 1, (12)	6094 77590 16096 85726 12782 98
1071	32; 1, 18, 1, (6); 46, 17, 35, 1, 38, (9)	90138 48491
1072	32; 1, 6, 1, 1, 8, 1, 4, 1, 1, (3); 47, 17, 35, 1, 38, (9)	880 28799
1073*	32; 1, (9, 9); 48, 16, 44, 9, 32, 33, 7, 49, 12, 33, 31, (16)	1457 20107 81927
1074	32; 1, 23, 31, 30, 25, (2); 49, 15, 3, (7, 7)	47710 1385 45368
1075	32; 50, 3, 1, 2, 3, 10, 1, 1, 1, 2, 2, 6, (1); 51, 14, 1, 19, 6, 39, 25, 34, 21, 26, 9, (50)	1 06476 12729
1076	32; 51, 14, 1, (3, 39, 21); 52, 4, 2, 15, (16)	34 89425 16 51504
1077	32; 52, 4, 2, 15, (16); 53, 29, 4, 1, (20)	88675 410
1078	32; 53, 29, 4, 1, (20); 54, 4, 10, 1, 1, 2, 2, 4, 3, 1, (1)	74226 13449
1079	32; 54, 4, 10, 1, 1, 2, 2, 4, 3, 1, (1); 55, 1, 35, 29, (3)	22399 2357 16760
1080	32; 55, 1, 35, 29, (3); 56, 9, 5, (6)	1805 76876
1081	32; 7, 1, 4, 3, 1, 6, 1, 1, 5, 1, 3, 1, 12, 2, 1, 1, 1, 21, 3, 2, (2); 57, 8, 15, 16, 45, 9, 33, 32, 11, 27, 15, 48, 5, 24, 33, 25, 40, 3, 19, 24, (23)	97325 54 161
1082*	32; (2, 7, 1, 5); 58, 7, 26, 23, 22, (31, 31)	8918 12221 87280 2 93215 05610 17151 07648
1083	32; (9); 59, (6)	19615 12 38369

TABLE 1001 TO 1500—CONTINUED.

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
1084	$32; 1, 1, 1, 3, 1, 2, 1, 1, 6, 1, 1, 1, 12, 5, 1, 1, 1, 21, 3, 4, (16); 60, 5, 12, 25, 24, 17, 27, 9, 51, 8, 23, 36, 19, 40, 15, 37, 24, 41, 3, 20, 15, (4)$	8 17041 27029 09269 93200 269 00393 64201 05379 19999
1085	$32; 1, 15, (2); 61, 4, (31)$	544 17919
1086	$32; 1, (20); 62, (3)$	22 725
1087	$32; 1, (31); 63, (2)$	33 1088
1088	$32; (1); (64)$	1 33
1090*	$33; (66); (1)$	33 1090
1091	$33; (33); (2)$	33 2
1092	$33; (22); (3)$	727 22
1093*	$33; 16, (1, 1); 4, (33, 33)$	18018 545
1094	$33; 13, 4, 1, 1, 1, 5, 2, 1, 2, 3, 9, 6, 1, 1, (32); 5, 14, 37, 25, 38, 11, 23, 35, 22, 19, 7, 10, 31, 35, (2)$	45043 80474 67914 14 89854 05815 38085
1095	$33; (11); (6)$	364 11
1096	$33; 9, 2, 3, 1, (15); 7, 28, 15, 49, (4)$	1 19595 39 59299
1097*	$33; 8, 3, 1, 3, 2, 1, (1, 1); 8, 17, 41, 16, 23, 32, (29, 29)$	2106 34621 19276
1098	$33; 7, 2, 1, (6); 9, 22, 41, (9)$	3564 1 18097
1099	$33; 6, 1, 1, 1, 1, 2, 21, 1, 2, 1, 1, 6, 1, (3); 10, 37, 27, 30, 31, 25, 3, 46, 19, 30, 35, 9, 47, (14)$	4 80575 45715 199
1100	$33; (6); (11)$	6 199
1101	$33; 5, 1, 1, 16, (22); 12, 31, 35, 4, (3)$	7 32732 243 13015
1102	$33; 5, 10, 1, 6, (2); 13, 6, 53, 9, (29)$	3 42882 113 82443
1103	$33; 4, 1, 2, 1, 2, 3, 1, 1, 5, 2, 9, (33); 14, 41, 19, 37, 22, 17, 31, 34, 11, 29, 7, (2)$	7 06456 11145 234 62427 45024
1104	$33; 4, 2, (2); 15, 25, (23)$	234 7775
1105*	$33; 4, (7, 7); 16, (9, 9)$	857

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
		28488

TABLE 1001 TO 1500—CONTINUED.

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
1106	33; 1, 1, (7); 17, 7, 2, (8)	152 5055
1107	33; 3, 46, 23, 9, (33); 18, 3, 37, (16)	18295 19428
1108	33; 3, 49, 16, 13, 48, 9, 36, 29, 23, 21, 39, 16, 27, 12, 7, 21, (4); 19, 28, 3, 4, 1, 6, 1, 1, 2, 2, 1, 3, 2, 5, 9, 3, (16)	63026 4781 20058 69390 13510
1109*	33; 3, 11, 44, 3, 13, 16, 1, 1, 2, 1, 4, (2, 2); 20, 3, 19, 5, 1, 17, 5, 4, 37, 29, 20, 41, 13, (25, 25)	35957 38617 1832 42610
1110	33; 3, (6); 21, (10)	61020 60015 60
1111	33; (3); (22)	1003 30
1112	33; 2, (7); 23, (8)	21056 2501
1113	33; 2, 17, 16, 47, (7); 24, 41, 2, 3, 3, 1, (8)	75 7 02463
1114*	33; 7, 55, 6, 15, 22, 39, 15, 42, 17, 9, 10, (33, 33); 25, 37, 2, 39, 8, 1, 10, 4, 2, 1, 3, 1, 3, 7, 6, (1, 1)	18311 57471 56745 6 11178 81032 02293
1115	33; 5, 2, 2, 2, 5, 1, (5); 26, 29, 3, 1, 4, (5)	45 1 36565
1116	33; 5, 1, 11, 40, 23, (4); 27, 25, 35, 11, 11, 49, (10)	46 60126 1 20799
1117*	33; 5, 1, 1, 7, 5, 4, 1, 21, 2, 9, 16, 1, 1, 1, 1, 6, 1, 4, 1, (2, 2); 28, 1, 2, 1, 5, 36, 23, 12, 13, 52, 3, 31, 7, 4, 39, 27, 28, 37, 9, 49, 12, 41, (21, 21)	38320 6272 59559 53855 43645
1118	33; 5, (2); 29, (26)	126 4213
1119	33; 5, 38, 25, 39, 10, (3); 30, 4, 1, 1, 1, 6, (22)	9 316 46364
1120	33; 5, (4); 31, 1, (15)	57255 1
1121	33; 9, 1, 1, 8, 1, 1, 1, 3, 1, 1, 7, 1, 4, 3, 1, (2); 32, 56, 7, 40, 25, 37, 16, 31, 35, 8, 49, 13, 17, 40, (19)	453 3765 15198 93975 58620
1122	33; (1); (33)	67

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
		51043 64951
1123	33; 4, 1, 21, 1, 5, 7, 3, 1, 1, 2, 2, 10, 1, 3, (33); 34, 33, 3, 54, 11, 9, 18, 33, 31, 22, 27, 6, 47, 17, (2)	49257 14799 16 50664 55626 32482
1124	33; 9, 13, 3, 3, 1, 1, 1, 1, 1, 2, (16); 35, 7, 5, 20, 17, 35, 28, 31, 29, 32, 25, (4)	2266 74757 22006 30049
1125	33; 11, 36, 5, 1, 29, 25, 9, 4, 41, 25, 36, 19, 44, (9); 36, 31, 2718, 91205, 173 04001	5 16002 173 04001
1126	33; 3, 1, (32); 37, 30, (2)	2718 91205

TABLE 1001 TO 1500—CONTINUED.

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
1127	33; 1, 3, (33); 38, 29, 19, (2)	1645 55224
1128	33; 1, 1, 4, (3); 39, 28, 13, (24)	70 2351
1129*	33; 1, 1, (1); 40, 27, (37)	5 168
1130*	33; 1, 1, (1, 1); 41, 25, (31, 31)	13 437
1131	33; 1, 1, 5, 2, 1, 1, 1, (4); 42, 13, 23, 35, 30, 35, (13)	10948 3 68185
1132	33; 1, 1, 1, 4, 1, 16, 3, 1, 1, 21, 1, 7, 2, 5, 7, 3, 2, 2, (16); 43, 24, 1, 23, 31, 36, 3, 57, 8, 29, 12, 9, 19, 24, 27, (4)	1 13654 94862 36362 38 23944 35058 85447
1133	33; 1, 23, 39, 13, 43, (6); 44, 2, (11)	107 14999 15300
1134	33; 1, 22, 4, 7, 2, 2, (4); 45, 21, 25, 9, 29, 26, 25, (14)	52041 3620 74049
1135	33; 1, 4, 10, 1, 1, 3, 1, 1, 2, 1, 1, 1, 6, 1, (5); 46, 20, 1, 23, 6, 49, 15, 34, 31, 21, 35, 26, 39, 9, 51, (10)	31 50736 19505 1061 47549
1136	33; 1, 14, 1, 2, 9, (4); 47, 19, 32, 31, 25, 7, (16)	56124 2427 18620
1137	33; 1, 3, 2, 1, 1, 7, 1, 5, 4, (22); 48, 18, 17, 17, 24, 29, 37, 8, 51, 11, 16, (3)	20799 26292
1138*	33; 4, 1, (1, 1); 49, 41, 14, (33, 33)	10337 3 48711

a	continued-fraction expansion of $\sqrt{a}$; period and subsidiary numbers	x y
1139	$33; (1); \quad (17)$	1354 50 16, (15)
1140	$33; (15); \quad (4)$	2431 51
1141	$33; 3, 1, 1, 15, 31, 36, 1, 12, 1, 21, 1, 1, 2, 1, 4, 3, 7, 5, 16, 1, 2, 3, 1, 1, 1, 2, 1, 3, 1, 1, 8, 5, 3, 2, 2, 7, 6, 5, 6, 5, 7, 19, 7, 52, 5, 60, 3, 39, 28, 25, 12, 15, 20, 9, 13, 4, 45, 21, 17, 36,$	3461988) 85322 76565 71973 (7) 103 67823 94157 22396 32371 97208
1142	$33; 1, 14, 11, 2, 1, 8, 1, (32); \quad 53, 13, 47, 1, 22, 43, 7, 59, (2)$	268 25215 9066 12101
1143	$33; 1, 3, 14, 5, 1, 1, 2, (3); \quad 54, 27, 34, 23, (18)$	47 28010 72
1144	$33; 12, 4, 4, 25, 1, 37, 1, 6, (1); \quad 55, 39, 40, 9, (52)$	1 39925 37
1145*	$33; 4, (5, 5); \quad 56, (15, 15)$	30624 1252
1146	$33; 10, 47, 5, 1, 1, 1, (10); \quad 57, 1, 38, 25, 41, (6)$	16611 5 61835